

SUBHASH MOTHUKURU

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 5 years of experience building and deploying production-scale machine learning, recommendation, and Generative AI systems. Experienced in developing ranking and personalization models, Retrieval-Augmented Generation (RAG) pipelines, agentic AI workflows, and LLM applications using Python, PyTorch, Hugging Face, LangChain, and LangGraph. Strong background in model fine-tuning, semantic retrieval, distributed ML pipelines, MLOps/LLMOps, and low-latency inference across AWS, Azure, and GCP. Proven ability to improve model performance, optimize production AI systems, and collaborate with product, engineering, and research teams to deliver scalable, measurable AI solutions.

PROFESSIONAL EXPERIENCE

AI/ML Engineer Scale AI

Jul 2024 – Present | San Francisco, CA

- Design and ship agentic AI workflows using LangChain and LangGraph, decomposing multi-step enterprise tasks into tool-calling and function-calling chains with Model Context Protocol (MCP) integrations, grounded retrieval, and structured fallback handling.
- Fine-tune transformer models (GPT, BERT, LLaMA 3) using LoRA, QLoRA, and PEFT for classification, summarization, and sentiment tasks, raising accuracy up to 18% over baseline on internal evaluation sets.
- Apply RLHF, prompt engineering, and human-in-the-loop feedback loops to align model behavior with product requirements, running structured offline evaluations to confirm gains before rollout.
- Build Retrieval-Augmented Generation (RAG) pipelines over Pinecone, Weaviate, FAISS, and ChromaDB, tuning chunking, embedding generation, hybrid search, and semantic reranking to improve answer relevance across 5+ enterprise products.
- Run LLM evaluation with RAGAS, DeepEval, and LangSmith, measuring groundedness, hallucination rate, and citation accuracy to gate releases and enforce AI guardrails before production rollout.
- Develop generative AI applications including LLM-powered chatbots, automated document summarization, and content moderation services, reducing manual review effort for enterprise workflow teams.
- Own end-to-end MLOps and LLMOps pipelines in MLflow, Kubeflow, Docker, and Kubernetes across Azure Machine Learning and GCP Vertex AI, standardizing experiment tracking, model versioning, drift detection, and automated retraining.
- Optimize low-latency inference with ONNX, TensorRT, quantization, vLLM, and Ray Serve behind FastAPI microservices, reducing serving cost roughly 28% while holding latency targets.
- Automate training and data ingestion pipelines with Python, PyTorch, Hugging Face, Pandas, NumPy, PySpark, and Airflow, cutting model deployment turnaround by 35% on multi-terabyte unstructured datasets.
- Partner with product managers and research engineers to translate ambiguous requirements into measurable evaluation criteria, deployment plans, access-control policies, and responsible AI governance standards.

ML Engineer Meta

Jun 2020 – May 2023 | Hyderabad, India

- Fine-tuned transformer-based text and image encoders in PyTorch to learn ad creative embeddings, replacing hand-engineered content features and improving click-through rate prediction on assigned ad surfaces.
- Engineered two-tower candidate retrieval and neural ranking models over sparse user and advertiser embeddings, combining collaborative filtering and content-based signals with negative sampling and probability calibration.
- Improved cold-start recommendation performance through content-based feature backfill and behavioral analytics on new advertiser accounts, validated offline with NDCG, Recall@K, Precision@K, and MAP before online rollout.
- Scaled model training with distributed data-parallel GPU workflows, mixed-precision, hyperparameter optimization, and feature selection, increasing the number of ranking experiments the team could run each week.
- Optimized low-latency model serving through ONNX export, TorchScript tracing, and int8 quantization, holding p99 latency under 100ms while serving 10M+ daily prediction requests on AWS (EC2, S3, Lambda, CloudWatch).
- Ran canary rollouts and online A/B tests through FBLeaer Flow, defining guardrail metrics and validating 5%+ incremental uplift in ad performance before full release.
- Automated retraining in CI/CD (GitHub Actions, Jenkins, AWS S3) with model versioning and drift detection, triggering refreshes from embedding-drift and calibration alerts instead of fixed schedules.
- Built petabyte-scale batch and real-time streaming pipelines in Spark SQL, PySpark, Presto, Hive, Kafka, and Scuba, consolidating fragmented ad interaction logs into a reusable feature store for modeling and BI teams.
- Partnered with Product, Data Science, and Responsible AI on brand-safety and content-classification filters, applying explainable AI (SHAP, LIME) and human-in-the-loop review to reduce policy-violating impressions without material recall loss.
- Collaborated across Agile sprint cycles through design reviews, code reviews, and technical documentation, onboarding new engineers onto ranking pipelines and driving continuous improvement of modeling standards.

TECHNICAL SKILLS

Programming & ML Frameworks: Python, SQL, Java, Bash, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Pandas, NumPy

AI, Machine Learning & Deep Learning: Generative AI, Machine Learning, Deep Learning, NLP, Computer Vision, Transformers, Supervised/Unsupervised Learning, Reinforcement Learning (RL/RLHF), Transfer Learning, Feature Engineering, Hyperparameter Optimization, Distributed GPU Training, Multimodal AI

Generative AI & LLMs: GPT, BERT, LLaMA, RAG, Agentic AI, AI Agents, LangChain, LangGraph, Model Context Protocol (MCP), Prompt Engineering, LoRA/QLoRA, PEFT, Tool/Function Calling, Human-in-the-Loop, AI Guardrails

Recommendation, Ranking & Retrieval: Two-Tower Networks, Candidate Retrieval, Learning-to-Rank, Collaborative Filtering, Personalization, CTR Prediction, Embeddings, Semantic/Hybrid Search, Pinecone, Weaviate, FAISS, ChromaDB, NDCG, Recall@K, Precision@K, A/B Testing

MLOps, LLMops & Evaluation: MLflow, Kubeflow, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Model Monitoring, Drift Detection, Experiment Tracking, RAGAS, DeepEval, LangSmith, ONNX, TensorRT, vLLM, Ray Serve, Quantization

Data Engineering & Cloud: Apache Spark, PySpark, Spark SQL, Airflow, Kafka, Hive, Presto, ETL/ELT, Feature Stores, AWS (SageMaker, EC2, S3, Lambda, CloudWatch), Azure ML, Azure Databricks, GCP Vertex AI

AI Engineering & Development: FastAPI, REST APIs, Microservices, Model Serving, Low-Latency Inference, SHAP, LIME, Explainable AI (XAI), Responsible AI, Bias/Fairness Evaluation, Git, System Design, Agile/Scrum

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals | AWS Educate: Introduction to Generative AI
- DeepLearning.AI: LangChain for LLM Application Development, AI Agents in LangGraph, Building Ambient Agents with LangGraph, Building AI Voice Agents for Production
- Fine-Tuning Language Models with Azure AI Foundry